Mechanism Analysis

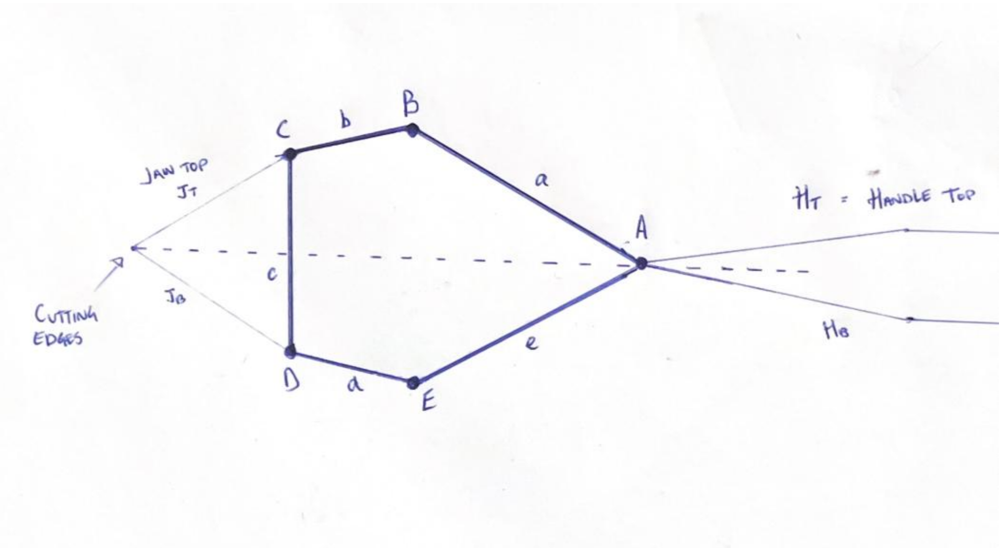
Knipex Bolt Cutters (71 72 460)

Personal experience

I have decided to look at the Knipex Bolt Cutters (71 72 460). I have used bolt cutters a fair amount from when I worked as a fabricator welder for cropping threaded rod, cutting thick gauge mesh, removing old concrete bolt stubs when an angle grinder is a fire risk etc. What I have noticed is that the input required although still sometimes difficult for one person to apply to two handles is considerably advantaged at the cutting jaws allowing you to cut through large sections of steel with just mechanical input. I have found from use on larger rebar that gripping at the ends of the handles and applying constant force is best for the mechanical advantage.

The handles on the bolt cutters are not straight. The handles are spread away from the centre line and then straighten out. The cutting jaws are closed when the angle between the handles at the point they spread apart is about 10 degrees and when the tool is fully open, they are about 100 degrees apart.

Kinematic diagram, links, pairs, and mobility



The Bolt Cutter is symmetric about the center line. There are five links: two cutting jaws, two handles, and one short link joining the jaws. These links are all connected by five revolute pins (1-DOF full joints, J_1). There are no sliders or higher pairs.

Kutzbach/Grübler Mobility

Given $L = 5, J_1 = 5$, and $J_2 = 0$:

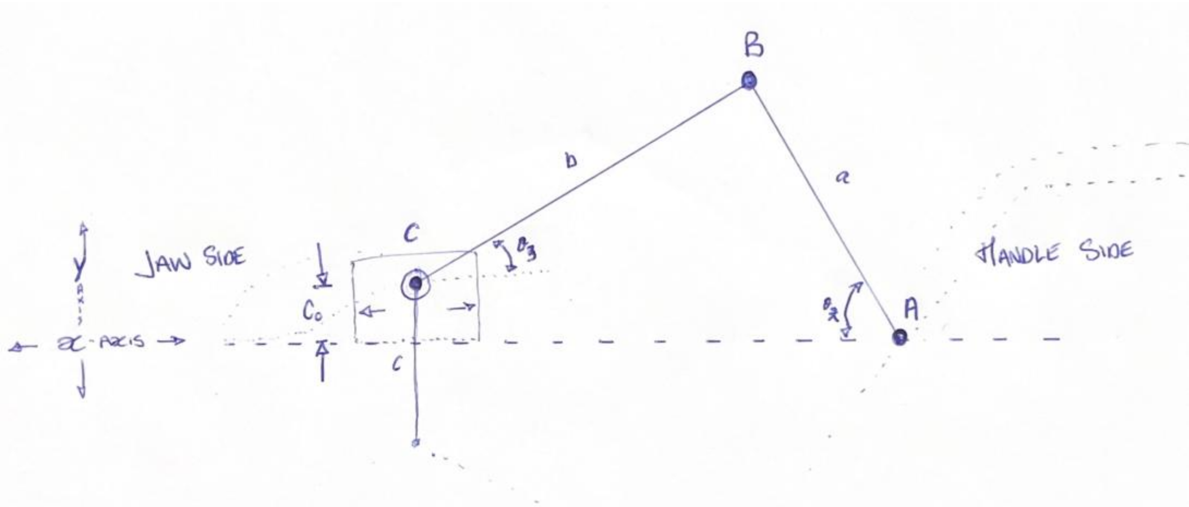
$$M = 3(5 - 1) - 2(5) = 2$$

The intrinsic mobility is two. Squeezing both handles together is how the tool is used, which practically ties the two inputs together during normal operation.

Ground, Input, Coupler, and Output

The cutter is a closed five-bar loop, and there is no bolted-in link locked to create a ground. Instead, the ground is the center line or axis of symmetry, which pin A lies on. From an analysis of the tool in the open and closed positions, it is clear that the tool forces itself to be symmetrical and that the short connecting link travels only along the x-axis (along the symmetry line). Hence, pin C (and D) maintains a constant offset (c_0) from that line.

The Offset Slider-Crank (Half-Model)



Because the tool is perfectly symmetric and driven evenly by both hands, the central cut line (the x-axis) acts as a fixed reference plane. This allows us to mathematically slice the closed 5-bar mechanism in half, simplifying it into a 1-DOF offset slider-crank mechanism.

On this half-model:

- The inner handle arm (a) acts as the crank, rotating about the ground pin A.
- The jaw arm (b) acts as the connecting rod from pin B to pin C.
- Jaw pin C acts as a slider, constrained to move purely horizontally at a fixed perpendicular offset (c_o) from the cut line.

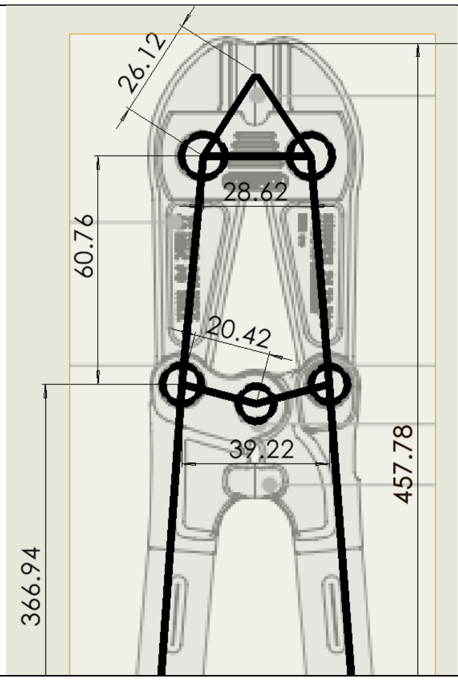
Component Breakdown

- Input: Motion is initiated via the handles. The full tool uses both handles acting together like a single input, despite the mechanism's mobility of 2. In the offset slider-crank half-model, the upper handle is isolated, and we focus on the inner portion length (pin A to pin B) driven by the angle θ_2. The extended grip (H_T) is the lever where force is applied. Because of the handle's profile, squeezing the grips towards the center line forces the inner arm (link a) to rotate toward the center line. The physical stroke limit assumptions I am making are for link a to be fully open at 40° and fully closed at 75°.

- Coupler: The coupler transfers motion from the handle to the jaw. In the half-model, this is link b, which acts as the connecting rod in an offset crank-slider mechanism. In the full tool, the addition of the short connecting link couples the top and bottom jaws, ensuring a synchronized movement.
- Output: The output is the cutting motion of the jaws. This is represented by the jaw angle (θ_3) at pin B. The cutting stroke is generated by this small angular rotation combined with the blade's offset (J_T) from pin C.

Parameters used for Analysis

Parameter	**Symbol**	**Value**
Handle inner arm (link a), A–B	a	20 mm
Jaw arm (link b), B–C	b	60 mm
Connecting link (link c), C–D	L_3	30 mm
Pin C offset	c_0	15 mm
Jaw lever (top)	J_T	$20 - 50$ mm ($\approx$ 25 nominal)
Handle lever (top), from A	H_T	$320 - 370$ mm ($\approx$ 340 typical)
Bottom jaw–handle pin spacing, B–E		40 mm
Pin-row separation (top to bottom)		60 mm
Overall tool length	L_{tool}	460 mm
Input handle angular speed	ω_h	0.25 rad/s



I used the technical data sheet to roughly (~ 3mm) get links, cutting edge, and handle lengths

Position analysis

Using the offset crank-slider and taking angles perpendicular to the center line:

$$a\sin\theta_2 + b\sin\theta_3 = \mathrm{c}_o \quad \Longrightarrow \quad \theta_3 = \arcsin\left(\frac{\mathrm{c}_o - a\sin\theta_2}{b}\right)$$

As the tool is squeezed, $\theta_2$increases, driving pin B upwards. At a mid-stroke input of $\theta_2 = 57.5°$:

$$\theta_3 = \arcsin\left(\frac{15 - 20\sin(57.5°)}{60}\right) = \arcsin(-0.031) = -1.784°$$

As Link a approaches its maximum angle 75°, pin B is driven slightly higher than the 15 mm offset, forcing the jaw link to angle slightly downwards making angle negative

The jaw angle flips from positive to negative. Pin B vertically passes the 15 mm offset pin C, forcing the jaw link to become perfectly horizontal ($\theta_3 = 0°$) mid-cut. Because $\cos(0°) = 1$, the kinematic multiplier(below) maxes. *This I believe is designed for the transition of when your stance changes from pulling the handles towards your core to when you can use your shoulders and core to press the handles together towards your chest where I believe you have the best force excursion.*

Velocity analysis

Differentiating the constraint with respect to time ($\omega_2 = \dot{\theta}_2, \omega_3 = \dot{\theta}_3$):

$$a\omega_2\cos\theta_2 + b\omega_3\cos\theta_3 = 0 \quad \Longrightarrow \quad \omega_3 = -\left(\frac{a\cos\theta_2}{b\cos\theta_3}\right)\omega_2$$

Assuming a steady input squeeze of $\omega_2 = +0.25$ rad/s at mid-stroke ($\theta_2 = 57.5°$):

$$\omega_3 = -\left(\frac{20\cos(57.5°)}{60\cos(-1.8°)}\right) = -0.045 \text{ rad/s}$$

The negative sign confirms the physical motion that as the inner handle arm rotates upwards, the jaw arm rotates downwards to close the cutting blades.

Ideal force ratio

Additional analysis

Assuming rigid links and no friction, power in equals power out. Moving the handles in at $H_T\omega_2$and the cutting material load in the cutting jaws at $J_T|\omega_3|$:

$$MA = \frac{F_{\text{jaw}}}{F_{\text{handle}}} = \left(\frac{H_T}{J_T}\right)\left|\frac{\omega_2}{\omega_3}\right| = \left(\frac{H_T}{J_T}\right)\left(\frac{b\cos\theta_3}{a\cos\theta_2}\right)$$

In the bolt cutters there are 2 levers working to create a multiplier at the cost of speed on the jaws side. The static lever based on the users input ($H_T/J_T \approx 9.71$) and the dynamic kinematic multiplier depending on input angle.

At mid-stroke ($\theta_2 = 57.5°$):

$$MA = \left(\frac{340}{35}\right)\left(\frac{60\cos(-1.8°)}{20\cos(57.5°)}\right) \approx 9.71 \times 5.58 \approx 54$$

This yields roughly a 54x force gain mid-cut. By the time the tool is fully closed at $\theta_2 = 75°$, the force ratio gains to $\approx 100 +$

Textbook problems

2-22

Find the Grashof condition and Barker classifications of the mechanisms in Figure P24a, b, and d.

2-22

a) $S + L \leq P + Q$

L_1 = GROUND = 174 = L

L_2 = INPUT = 116 = P

L_3 = COUPLER = 108 = S

L_4 = OUTPUT = 110 = Q

108 + 174 = 282

116 + 110 = 226

282 > 226 = NON GRASHOF (RRR) TRIPLE ROCKER. CLASS II

b) L_1 = 162 = GROUND = L

L_2 = 40 = INPUT = S

L_3 = 122 = COUPLER = P

L_4 = 96 = OUTPUT = Q

162 + 40 = 202

122 + 96 = 218

202 < 218 = GRASHOF CRANK ROCKER (GCRR) CLASS)

d) L_1 = 150 = GROUND

L_2 = 30 = INPUT

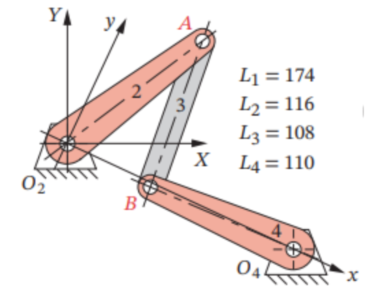
L_3 = 150 = COUPLER

L_4 = 30 = OUTPUT

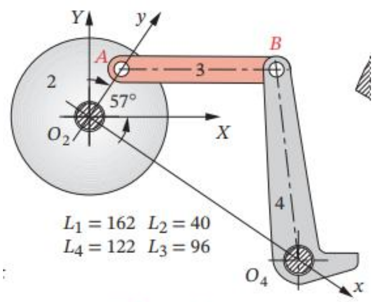
150 + 30 = 180

150 + 30 = 180

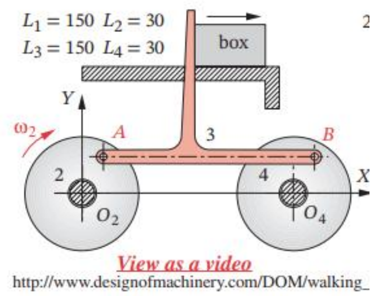
180 = 180 = SPECIAL GRASHOF PARALLELOGRAM DOUBLE CRANK - CLASS III



(a) Fourbar linkage



(b) Fourbar linkage



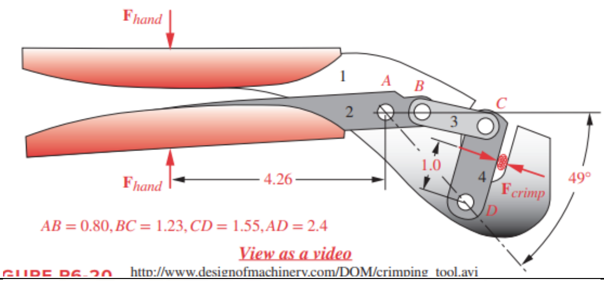
(d) Walking-beam conveyor

6-57

Figure P6-20 shows a crimping tool.

a. Calculate its mechanical advantage for the position shown.

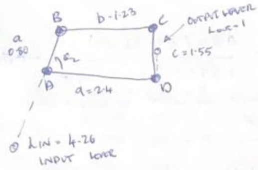
b. Calculate and plot its mechanical advantage as a function of the angle of link AB as it rotates from 60 to 45°.



GURE P6-20

6-57

a



POSITION ANALYSIS.

$K_1 = \frac{d}{a} = \frac{2.4}{0.8} = 3$

$K_2 = \frac{d}{c} = \frac{2.4}{1.55} = 1.548$

$K_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac}$

$= \frac{0.8^2 - 1.23^2 + 1.55^2 + 2.4^2}{2(0.8)(1.55)}$

$= \frac{7.289}{2.48} = 2.9$

$A = \cos\theta_2 - K_1 - K_2\cos\theta_2 + K_3$

$= \cos(49) - 3 - 1.548\cos 49 + 2.939$

$= -0.420.$

$B = -2\sin\theta_2 = -2\sin(49) = -1.509$

$C = K_1 - (K_2 + 1)\cos\theta_2 + K_3$

$= 3 - 2.548\cos(49) + 2.939$

$= 4.67$

$\theta_4 = 2\tan^{-1}\left(\frac{-B - \sqrt{B^2 - 4AC}}{2A}\right)$

$= 2\tan^{-1}\left(\frac{1.509 - 3.075}{-0.840}\right)$

$= 123.5°$

TO FIND θ_3:

$\sin\theta_3 = \frac{c\sin\theta_4 - a\sin\theta_2}{b} = \frac{1.55\sin(123.5°) - 0.8\sin(49)}{1.23} = 0.560$

$\cos\theta_3 = \frac{d + c\cos\theta_4 - a\cos\theta_2}{b} = \frac{2.4 + 1.55\cos(123.5) - 0.8\cos(49)}{1.23} = 0.828$

$\theta_3 = \tan^{-1}\left(\frac{0.560}{0.828}\right) = 34.07.$

6-57 CONTINUED..

VELOCITY ANALYSIS.

$\frac{\omega_4}{\omega_2} = \frac{a\sin(\theta_2 - \theta_3)}{c\sin(\theta_4 - \theta_3)} = \frac{0.8\sin(49 - 34.07)}{1.55\sin(123.5 - 34.07)}$ = 0.133

$= \frac{0.207}{1.55} = 0.133$ RATIO OF INPUT TO OUTPUT SPEED.

MECHANICAL ADVANTAGE

$MA = \frac{F_{CRIMP}}{F_{HAND}} = \frac{l_{IN}}{l_{OUT}}\left|\frac{\omega_2}{\omega_4}\right|$

MA at 49°:

$MA = \left(\frac{4.26}{1.0}\right)\left(\frac{1}{0.133}\right) = 32$

MA at $\theta_2 = 60°$:

REPEAT OF ABOVE, USING CALCULATOR:

$A = -0.335$, $B = -1.732$, $C = 4.665$

$\theta_4 = 125.8°$ $\theta_3 = 27.3°$ $\frac{\omega_4}{\omega_2} = 0.281$

$MA = \left(\frac{4.26}{1}\right)\left|\frac{1}{0.281}\right| = 15.1$

MA at $\theta_2 = 45°$:

$A = -0.449$, $B = -1.414$, $C = 4.137$

$\theta_4 = 123.41°$ $\theta_3 = 36.6°$ $\frac{\omega_4}{\omega_2} = 0.076$

$MA = \left(\frac{4.26}{1}\right)\left|\frac{1}{0.076}\right| = 56.1$

2-31

Figure P2-12 shows a bicycle brake caliper assembly. Sketch a kinematic diagram of this device and draw its equivalent linkage. Determine its mobility under two condi- tions:

a. Brake pads not contacting the wheel rim.

b. Brake pads contacting the wheel rim. Hint: Consider the flexible cables to be replaced by forces in this case.

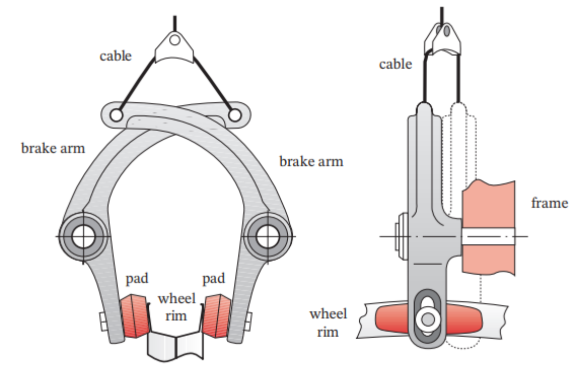
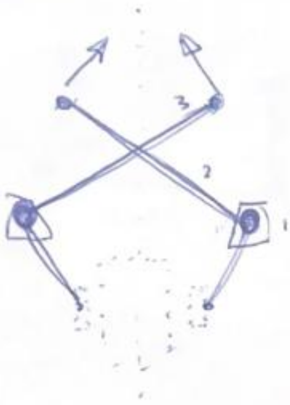


FIGURE P2-12

Problem 2-31 Bicycle brake caliper assembly

2-31

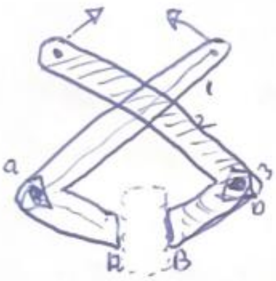


$L = 3 \quad J_1 = 2$

$M = 3(3-1) - 2(2) - 0 =$

$6 - 4 =$

2. MOBILITY. WHEN NOT CONTACTING WHEEL RIM - BOTH ARMS CAN ACT INDEPENDANTLY.



$L(1,2,3) = 3$

$J_1(a,b) = 2$

$J_2(A,B) = 2$

$M = 3(3-1) - 2(2) - 1(2) =$

$6 - 4 - 2 = 0$

MOBILITY GOES TO ZERO AND TRANSFERS FORCE INTO THE RIM.

References

Norton, R. L., *Design of Machinery*, 6th ed., McGraw-Hill

KNIPEX, *Operating Instructions found on https://www.knipex.com/products/cutting-pliers/bolt-cutters/bolt-cutters/7172460*